Devan Manoj

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Profile

Results-driven Software Development Engineer with hands-on experience building scalable backend systems and AI-powered platforms. Skilled in designing microservices, APIs, and integrating large language models into production workflows. Strong background in performance tuning, cloud deployment, and search/indexing systems, with proven ability to improve SLA compliance, scalability, and response quality in GenAI-driven applications.

Professional Experience

Software Engineer I 05/2024 – present

Thermo Fisher Scientific

- Designed and deployed scalable backend services for AI-driven image and video generation pipelines, supporting production GenAI content workflows.
- Built agent-based hallucination validation workflows to improve generated content reliability and production readiness.
- Enhanced ChatGPT-based workflow authorization, orchestration, and API integrations, improving system security and response consistency.
- Reduced response latency and improved SLA adherence by scaling cloud infrastructure and refactoring orchestration logic.
- Engineered high-throughput image-processing pipelines with optimised API endpoints, enabling automated PDP content workflows.
- Architected a user management microservice enabling agent- and tool-level access control within a microservices ecosystem, improving modularity and platform security.
- Executed JMeter-based load testing for up to 10,000 concurrent users, identifying bottlenecks and validating system scalability, stability, and fault tolerance under peak traffic.
- Migrated structured JSON data from blob storage to Cosmos DB while ensuring data integrity, and deployed production-grade AWS infrastructure to resolve deployment blockers and improve operational reliability.
- Integrated Azure Search Skills for language detection, translation, image analysis, and document extraction, and implemented predictive keyphrase extraction and citation features, improving multilingual retrieval accuracy, search relevance, and user trust.

Projects

AI-Based Product Image Generation & Validation System

- Built an automated backend system for product image generation with agent-based quality validation to detect hallucinations.
- Developed scalable processing pipelines supporting high-throughput (3K) image outputs.
- Built a UI enabling users to test the pipeline on individual images and iteratively refine outputs to explore the tool's full range of capabilities.
- Improved inference reliability through orchestration enhancements and structured validation workflows.

LLM-Integrated Video Generation Pipeline

- Refined video generation pipeline and evaluated LLM integration strategies to improve orchestration reliability, generation consistency, and end-to-end latency.
- Automated AI video creation from image URLs using SORA video model, enabling streamlined multimodal content generation workflows.
- Re-engineered the workflow using FFmpeg to achieve equivalent output quality at significantly lower cost and faster execution, with greater processing control and repeatability; adopted as the primary production approach.

Competitor Product Intelligence Pipeline

- Built a 5-phase pipeline to extract, structure, and internally surface competitor antibody product data, operating in compliance with source sites' robots.txt policies.
- Automated extraction of competitor product listings and full-page HTML capture, followed by targeted key-point extraction, cleaning, and normalisation.
- Ingested processed data into an internal database, enabling SKU-level competitor lookups matched against internal product catalog using antibody specific attributes.
- Delivered a queryable interface allowing internal teams to retrieve competitor SKU details alongside corresponding matched internal products to support competitive analysis.

Education

Amrita Vishwa Vidyapeetham

Bachelor of Technology: Artificial Intelligence Engineering *09/2020 – 05/2024*

Skills

Backend: FastAPI, REST APIs, Microservices, API Design

AI/ML: LLMs, RAG pipelines, Embeddings, Vector Search, Agentic & tool-augmented workflows, Prompt Engineering, OpenAI / Azure AI integrations

Cloud & Infrastructure: AWS, Microsoft Azure, Docker, Kubernetes, Terraform, CI/CD

Databases & Search: Cosmos DB, AWS RDS, Azure AI Search

Tools: Git, JMeter, FFmpeg

Soft Skills: Adaptability and problem-solving; Ethical reasoning in AI; Collaboration between engineering and product teams; Business-aligned engineering outcomes